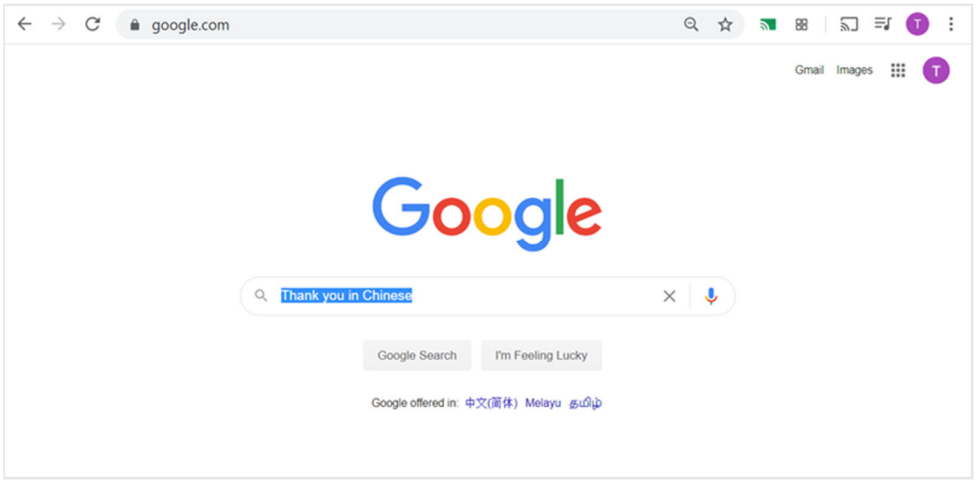
The student uses the Google Search form to translate “Thank you” into the Chinese characters “谢谢”. The Unicode character “谢” was previously encoded using the UTF-16 format, as U+8C22, before transmitting over the internet. But this has been replaced by the widely accepted UTF-8 encoding format.



(e) Explain how the web server will use server-side script to process this form. [4]

Upon receiving the input from the client, the server-side script will:

1. Check that it is not a null or empty input
2. Split the sentence into words, by the spacing in between
3. Check each word against the database for the corresponding translation
4. Compose the translated message and render a webpage to reply the client

(f) Convert the 16 bits hexadecimal 8C22 to its binary representation so that it can be transmitted over the internet. [2]

1	0	0	0	1	1	0	0	0	0	1	0	0	0	1	0

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(g) Explain why the UTF-16 has been replaced by UTF-8 encoding? [2]

[1] UTF-8 is backward compatible to ASCII.

[1] UTF-8 can be used to encode 2^21 different characters as compared to 2^16 for UTF-16.

Number of bytes	Bits for code point	First code point	Last code point	Byte 1	Byte 2	Byte 3	Byte 4
1	7	U+0000	U+007F	0xxxxxxx			
2	11	U+0080	U+07FF	110xxxxx	10xxxxxx		
3	16	U+0800	U+FFFF	1110xxxx	10xxxxxx	10xxxxxx	
4	21	U+10000	U+10FFFF[18]	11110xxx	10xxxxxx	10xxxxxx	10xxxxxx

(h) With the UTF-8 encoding, the data can only occupy the vacancies in the Leading and Continuation Bytes as shown in the following:

1	1	1	0	1	0	0	0

Leading Byte

1	0	1	1	0	0	0	0

Continuation Btye

1	0	1	0	0	0	1	0

Continuation Btye

State the three btyes, in hexadecimal, which will be transmitted over the internet for the Unicode character “谢”. [2]

E8 B0 A2

[1] At least one of the 3 bytes is correct.

[2] All 3 bytes correct

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